



The Signaling Gap

State-Owned Media and the Strategic Management of Sentiment during the Russian Invasion of Ukraine

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Abstract

Alliance commitments enable cooperation but can foreclose dissent: a state bound to a patron cannot openly break with it without risking its security guarantee. We argue state-owned media is an institutionally mediated channel — public enough for the foreign governments that monitor it to read, but not a formal act of state.

Across 174,094 articles from 11 countries around the 2022 invasion of Ukraine, states with Russian security agreements shifted **more negative** toward Russia than neutral states — even while refusing to condemn it through formal channels. We measure sentiment with a prompted large language model and identify the shift with a regression discontinuity.

The Argument

The bind. A state that depends on a patron for its security can't openly break with it without risking the guarantee it relies on — yet silence risks being lumped in with the patron.

The channel we test: state media. The state owns these outlets, so their coverage is a controlled choice, not an independent press responding to the market.

Why it works as a signal. *Readable* — foreign analysts monitor state outlets and treat them as regime-controlled, so a turn against Russia reads as a deliberate signal. *Costly* — negative coverage of a security patron risks provoking retaliation, so it's hard to fake.

The calculus. Signaling risks retaliation; silence risks being categorized with the aggressor. Because diplomatic isolation is the greater cost, sending the signal is rational — and the logic travels to any patron–client hierarchy.

The rival account. The dominant view treats state media as managing domestic opinion, not signaling outsiders — so a constrained state should mute criticism of its patron, not amplify it. The two accounts predict opposite shifts.

Hypotheses

H1 · Baseline shock — the invasion drove a region-wide negative shift toward Russia.

H2 · Readability — state media manages coverage more carefully than independent media.

H3 · The Signal — Alliance-constrained state media shifts more negative than neutral state media.

H3 is where the two accounts split:

Domestic management
Suppress coverage →
predicts less negative

Signaling (our account)
Signal via media → predicts
more negative

Data: shifts more negative ✓ signaling, not suppression

The Case: Why the 2022 Invasion

A clean test needs three things, and February 24, 2022 supplies all three:

- **A discrete shock** — a sharp rupture in the information environment, not a slow slide.
- **Alliance constraints** — states bound to Russia (CSTO + Azerbaijan) vs. neutral states with open channels.
- **Competing predictions** — domestic-management and signaling disagree on which group shifts more.

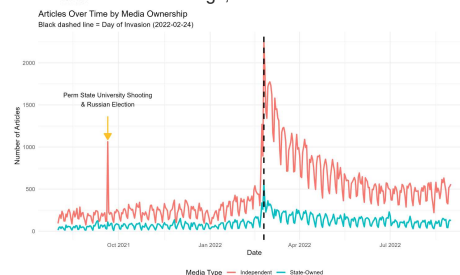
Russia-aligned: Armenia, Azerbaijan, Belarus, Kazakhstan

Neutral: Georgia, Kosovo, Moldova, Serbia

Data & Measurement

- **174,094** Russia-related articles, Aug 2021–Aug 2022; 11 countries, 61 sources (17 state-owned, 44 independent).

- **GPT-4o** codes sentiment toward Russia (−1 / 0 / +1) from each title and first two sentences. F1 = 0.92 vs. a 100-article human standard — with no −1 ↔ +1 cross-codings, a conservative measure.



Daily Russia-related articles by ownership; dashed line marks the Feb 24 invasion.

Methods Challenges

A credible discontinuity design here — ultimately an RDID across groups — must clear three obstacles first:

- **Composition, not tone.** The invasion changes what outlets cover, not just how. Topic is a collider, so we block topic, country, and ownership together via source-level control.
- **The collinearity bind.** Each outlet sits in one group, so source fixed effects would absorb the very contrast we test. We z-score within source instead — the interaction stays estimable.
- **Dates pile up.** Hundreds of articles share a date, creating mass points that break a time RDD. We use publication date-time as the running variable, imputing missing times per outlet.

Research Design

H1 (within group). An RD at Feb 24 identifies the causal effect of the invasion on sentiment (rdrobust, MSE-optimal bandwidth).

H2 / H3 (across groups). A difference-in-discontinuities (RDID) compares how much more one group jumped at the cutoff:

$$Y_{it}^s = \beta_0 + \beta_1 D_{it} + \beta_2 T_{it} + \beta_3 D_{it} T_{it} + \beta_4 G_{it} + \beta_5 T_{it} G_{it} + \beta_6 D_{it} G_{it} + \beta_7 D_{it} T_{it} G_{it} + Y_{itopic} + \epsilon_{it}$$

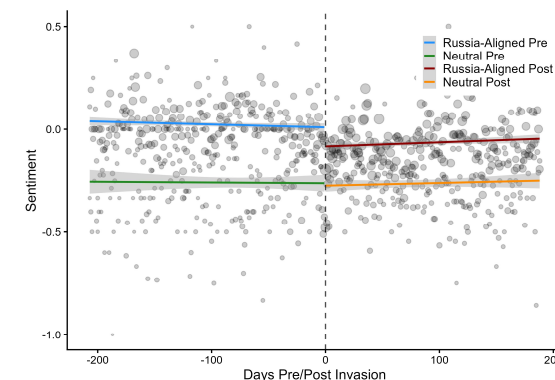
β_5 (T × G) is the differential discontinuity — causal within each group, but *conditional on group membership* across groups (stress-tested, not a causal effect of alignment).

Results

H1 — A real shock. Sentiment toward Russia fell 0.22 SD at the cutoff ($p < 0.001$); 0.18 SD without military coverage.

H2 — State media holds the line. Independent media fell 0.21 SD; state-owned only 0.11 SD. RDID interaction 0.22 ($p < 0.05$) among non-NATO outlets.

H3 — Constrained states signaled loudest. Russia-aligned state media fell −0.22 SD ($p < 0.05$); neutral barely moved (−0.06, n.s.). Matched estimate −0.38 ($p < 0.05$) — the opposite of the domestic-management prediction.



Russia-aligned drops sharply at the cutoff; neutral stays flat.

Robustness

Sensitivity (Cinelli–Hazlett). A confounder would need to be 1.5× as strongly associated with sentiment as the invasion itself to erase the H3 differential.

Placebo + ethnic partition. No discontinuity at the Dec 7 announcement; splitting by ethnic-Russian share yields no differential.

Few-cluster inference. With only ~13 clusters in the state-media test, a wild cluster restricted bootstrap (Webb weights) leaves the H3 differential intact.

Conclusion

- Connects patron–client theory and IR signaling — constrained clients use managed media to signal divergence from a patron.
- Controlled-media sentiment is a deliberate signal of alignment toward a security partner — not a byproduct of domestic opinion.
- Validates a long-standing intelligence practice — reading state media for regime preferences — long left unexamined.

Key References

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